b) In alkaline medium:

In the presence of alkali metal hydroxides, the permanganate ion is converted into manganate.

$$MnO_4^- + e^- \longrightarrow MnO_4^{2-}$$

This manganate is further reduced to MnO_2 by some reducing agents.

$$MnO_4^{2-} + 2H_2O + 2e^- \longrightarrow MnO_2 + 4OH^-$$

So the overall reaction can be written as follows.

$$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$$

This reaction is similar as that for neutral medium.

Baeyer's reagent:

Cold dilute alkaline $KMnO_4$ is known as Baeyer's reagent. It is used to oxidise alkenes into diols. For example, ethylene can be converted into ethylene glycol and this reaction is used as a test for unsaturation.

c) In acid medium:

In the presence of dilute sulphuric acid, potassium permanganate acts as a very strong oxidising agent. Permanganate ion is converted into Mn^{2+} ion.

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$$

The oxidising nature of potassium permanganate (permanganate ion) in acid medium is illustrated in the following examples.

(i) It oxidises ferrous salts to ferric salts.

$$2MnO_4^- + 10Fe^{2+} + 16H^+ \longrightarrow 2Mn^{2+} + 10Fe^{3+} + 8H_2O$$

(ii) It oxidises iodide ions to iodine

$$2MnO_4^- + 10\,I^- + 16H^+ \longrightarrow 2Mn^{2+} + 5I_2 + 8H_2O$$

(iii) It oxidises oxalic acid to CO_2

$$2MnO_4^- + 5\left(COO\right)_2^{2-} + 16H^+ \longrightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

(iv) It oxidises sulphide ion to sulphur

$$2MnO_4^- + 5\,S^{2-} + 16H^+ \longrightarrow 2Mn^{2+} + 5\,S + 8H_2O$$

(v) It oxidises nitrites to nitrates

$$2MnO_4^- + 5NO_2^- + 6H^+ \longrightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$$

(vi) It oxidises alcohols to aldehydes.

$$2KMnO_4 + 3H_2SO_4 + 5CH_3CH_2OH \longrightarrow K_2SO_4 + 2MnSO_4 + 5CH_3CHO + 8H_2O$$

(vii) It oxidises sulphite to sulphate

$$2MnO_4^- + 5SO_3^{2-} + 6H^+ \longrightarrow 2Mn^{2+} + 5SO_4^{2-} + 3H_2O$$

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